📘 Tutorial 4 — Add Timestamps

> Continue the React + Firestore Todo series by storing creation time for each task.

✅ What You Will Build

Each todo document will include a Firestore server timestamp:

- `createdAt: serverTimestamp()`

You will also load todos in newest-first order.

🎯 Learning Goals

By completing this tutorial you will:

- Use Firestore server-side timestamps
- Query data with `orderBy`
- Display formatted date/time in React

🧰 Before You Start

You should already have:

- Tutorial 3 completed
- A working `Todo.js` component loading todos from Firestore

Part 1 — Update Firestore Imports

Open:

```
src/components/Todo.js
```

Change Firestore imports from:

```
import { collection, addDoc, getDocs } from "firebase/firestore";
```

To:

```
```

```
import {
  collection,
  addDoc,
  getDocs,
  serverTimestamp,
  query,
  orderBy,
} from "firebase/firestore";
```

Part 2 — Save Timestamp on Create

Find `addDoc(...)` and update it to:

```
await addDoc(collection(db, "todos"), {
  todo,
  createdAt: serverTimestamp(),
});
```

Part 3 — Query Todos by Time

Replace your `fetchTodos` function with:

```
const fetchTodos = async () => {
  const q = query(collection(db, "todos"), orderBy("createdAt", "desc"));
  const snapshot = await getDocs(q);

  const list = snapshot.docs.map((doc) => ({
    id: doc.id,
    ...doc.data(),
  }));

  setTodos(list);
};
```

Part 4 — Show Timestamp in the UI

Update map output to include a readable date:

```
{todos.map((t) => (
  <div key={t.id} className="todo-row">
    <p>{t.todo}</p>

    <small className="time">
      {t.createdAt?.toDate
        ? t.createdAt.toDate().toLocaleString()
        : "Saving timestamp..."}
    </small>
  </div>
))}
```

Part 5 — Add Styles

Open:

```
src/App.css
```

Add:

```
.todo-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 1rem;
  margin-bottom: 0.75rem;
}

.time {
  color: #666;
  font-size: 0.8rem;
}
```

Part 6 — Test It

1. Add several todos
2. Confirm new ones appear first
3. Confirm each displays date/time

You should see:

✅ `createdAt` field in Firestore ✅ Newest items first ✅ Human-readable timestamps in UI

🛠️ Common Errors

❌ `orderBy` requires an index

Firestore may prompt for an index in advanced queries. Follow the console link if needed.

❌ Timestamp shows “Saving timestamp…” forever

Refresh once. Server timestamps can be null briefly right after write.

📚 Docs

- Server timestamps (Firestore)
- Order and limit data (Firestore)

➡️ Next: Tutorial 5 — Realtime Listeners